COMPUTER NETWORKS

Unit 1: Introduction • Unit 2: The Physical Layer

Topics Covered

1. Network Applications
2. Network Hardware
3. Network Software
4. Reference Models: OSI, TCP/IP, Internet, Frame Relay
5. Guided Transmission Media
6. Wireless Transmission
7. Public Switched Telephone Network (PSTN)
8. Mobile Telephone System

UNIT 1: INTRODUCTION TO COMPUTER NETWORKS

1.1 What Is a Computer Network?

A computer network is a collection of autonomous (independent) computers that are interconnected by a single technology so that they can exchange information and share resources such as printers, files, and internet connections. Two computers are said to be interconnected if they are able to exchange information with each other, using wires, fibre optics, microwaves, or any other medium.

Think of it like a city's road network: each house (computer) is connected by roads (cables/wireless links) so that people (data) can travel from one house to another. Without roads connecting them, the houses would be isolated — exactly like a computer with no network connection.

A network is different from a "distributed system", where the existence of multiple autonomous computers is transparent (hidden) to the user — the whole system looks like one single computer (for example, the World Wide Web).

Goals / Advantages of Computer Networks

- **Resource sharing:** many users can share one expensive resource, such as a printer or an internet connection, instead of buying one for every single computer.
- **High reliability:** if one machine or file copy fails, another connected machine can take over, so the work is not lost (this is why important files are often kept on more than one server).
- **Cost saving:** small, inexpensive computers can be networked together to do the job that used to need one very expensive large computer.
- **Scalability:** it is easy to add more users or computers to a network gradually, without redesigning everything.
- **Communication medium:** a network lets people who are far apart work together and exchange information instantly (e-mail, video calls, shared documents).

1.2 Uses of Computer Networks (Network Applications)

Computer networks are used for many purposes. The main categories are:

(a) Business Applications

- Resource sharing: sharing printers, scanners, files, and software among many users so the company does not need to buy separate copies for everyone.
- Client–server model: powerful server machines store data and provide services; client machines (desktops/laptops) request services from the server — much like customers (clients) ordering food from a restaurant kitchen (server).
- E-mail, video conferencing, and Voice over IP (VoIP — making phone calls by sending your voice as data packets over the Internet instead of the traditional telephone network) for business communication.
- E-commerce: online buying/selling, banking, and electronic data interchange (EDI — companies exchanging business documents like invoices and orders electronically instead of on paper).

(b) Home Applications

- Access to remote information: browsing the web, online news, online banking — all done by fetching information stored on a computer somewhere else in the world.
- Person-to-person communication: e-mail, social networking, video calls, instant messaging.
- Interactive entertainment: online gaming, streaming movies and music (OTT — Over-The-Top — services such as video/music streaming platforms).
- Electronic commerce: online shopping, digital payments — essentially a shop that is open 24 hours a day, reachable from home.

(c) Mobile Users

- Mobile phones and laptops connect to networks while on the move using Wi-Fi or cellular (mobile) data.
- Applications: mobile banking, GPS navigation, mobile e-mail, m-commerce (mobile commerce — shopping directly from a mobile phone).

(d) Social Issues

- Networks also raise social and legal issues — privacy of personal data, cybercrime, spam (unwanted bulk e-mail), fake news, and copyright violation (piracy — illegally copying and distributing protected content).

★ Quick Recap – 1.1 & 1.2

- A network connects independent computers so they can share resources and exchange data (like roads connecting houses in a city).
- Main goals: resource sharing, high reliability, cost saving, scalability, and communication.
- Main uses: business (resource sharing, e-commerce), home (entertainment, communication), mobile users, and related social issues.

1.3 Network Hardware

Network hardware can be classified in several useful ways: by the transmission technology used, by the scale (size) of the network, by its layout or "topology", and by the physical devices used to build it.

(a) Classification by Transmission Technology

- **Broadcast links:** a single communication channel is shared by all the machines on the network. Short messages called packets are sent by one machine and received by all others — like a radio station broadcasting to every listener tuned in. Usually used in small, localized networks (e.g., LAN, wireless networks).
- **Point-to-Point links:** a dedicated connection exists between one pair of machines, like a private telephone call between exactly two people. A packet may have to pass through several intermediate machines before reaching its destination (this is called unicasting). Usually used in large networks (e.g., WAN).

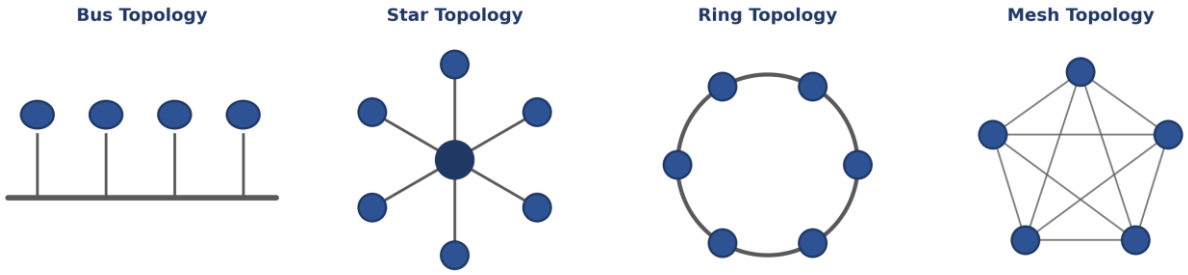
(b) Classification by Scale

Networks are also grouped by their physical size/distance covered, as shown below:

Network Type	Distance / Coverage	Example
PAN (Personal Area Network)	A few metres (around one person)	Bluetooth connecting phone, earphones, smartwatch
LAN (Local Area Network)	Up to a few kilometres (one building/campus)	Office network, college computer lab, home Wi-Fi
MAN (Metropolitan Area Network)	A city (tens of km)	Cable TV network, city-wide Wi-Fi
WAN (Wide Area Network)	A country or continent	The Internet, banking networks connecting branches across a country
Internetwork	Interconnection of many networks	The Internet itself — a "network of networks"

(c) Network Topologies (How Computers Are Physically Laid Out)

The topology of a network is the pattern or shape in which the computers (nodes) are connected to each other. Choosing a topology affects the network's cost, reliability, and how easy it is to add new machines.



The four basic network topologies

Topology	How It Works	Advantage	Disadvantage
Bus	All computers share one common central cable (the "bus")	Cheap and simple to install	If the main cable fails, the whole network goes down
Star	Every computer connects individually to one central device (hub/switch)	Easy to add/remove a computer; one failed cable doesn't affect others	If the central device fails, the whole network goes down
Ring	Each computer connects to exactly two neighbours, forming a closed loop	Data flows in an orderly manner, no collisions	A single broken link can disrupt the whole ring (unless a dual ring is used)

Mesh	Every computer is directly connected to every other computer	Very reliable — many alternate paths if one link fails	Expensive and complex to wire (needs a lot of cabling)

(d) Common Network Devices

These are the physical devices used to build and connect networks:

Device	Function (in simple terms)
Hub	A basic device that simply repeats/broadcasts incoming data to every connected computer — like shouting a message to an entire room even if it is meant for only one person.
Switch	A smarter version of a hub — it learns which computer is connected to which port and sends data only to the intended recipient, like a telephone operator connecting only the two people who want to talk.
Bridge	Connects and filters traffic between two LAN segments, forwarding only the data that needs to cross over, reducing unnecessary traffic.
Router	Connects two or more different networks and decides the best path to forward data between them — like a traffic police officer directing vehicles (data) toward the correct road (network).
Gateway	Connects networks that use completely different protocols, translating between them so they can communicate (e.g., connecting a company network to the Internet).
Modem	Converts (modulates) digital computer data into analog signals for transmission over telephone lines, and converts (demodulates) it back to digital at the receiving end.
Access Point (AP)	Allows wireless (Wi-Fi) devices to connect to a wired network.

★ Quick Recap – 1.3

- Broadcast = shared channel for all; Point-to-Point = dedicated link between two machines.
- Scale order (small to large): PAN → LAN → MAN → WAN → Internetwork.

- Topologies: Bus (one shared cable), Star (central device), Ring (closed loop), Mesh (every node to every node).
- Key devices: Hub (dumb broadcaster) < Switch (smart, targeted) < Router (connects different networks) < Gateway (translates protocols).

1.4 Network Software

(a) Protocol Hierarchies (Layers)

To reduce design complexity, most networks are organised as a series of layers or levels, each one built upon the one below it. This is similar to how a courier company works: you (the application) hand your parcel to the front counter; you don't need to know how the dispatch team routes it or which truck driver carries it — each level does its own job and passes the parcel down to the level below it.

- **Protocol:** an agreement between the communicating parties on how communication is to proceed — the rules and conventions used in a conversation between layer n on one machine and layer n on another machine (peers).
- **Interface:** defines which primitive operations and services the lower layer offers to the layer above it.
- **Peers:** the entities that occupy the same layer on different machines.
- **Architecture:** the set of layers and protocols is called the network architecture.

(b) Design Issues for the Layers

- **Addressing:** a mechanism to identify senders and receivers, just like every house needs a postal address and every phone needs a phone number.
- **Error Control:** agreeing on error-detecting and error-correcting codes so data arrives correctly — similar to counting the items in a shipment before and after delivery to check nothing is missing or damaged.
- **Flow Control:** keeping a fast sender from overwhelming a slow receiver — like a person dictating a letter too fast for the writer, so the writer occasionally says "wait, let me catch up".
- **Multiplexing/De-multiplexing:** sharing one connection among many independent conversations, the same way many radio stations share the airwaves using different frequencies.
- **Routing:** choosing the best path when there are multiple paths between source and destination — just like a GPS choosing the fastest road route.

(c) A Note on Error Detection and Flow Control Methods

- **Parity bit:** the simplest error-detection method — an extra bit is added so that the total number of 1-bits is always even (or always odd). If a single bit gets corrupted in transit, the count no longer matches and an error is detected.

- **Checksum / CRC (Cyclic Redundancy Check):** more powerful techniques where a small calculated value is sent along with the data; the receiver redoes the same calculation and compares the result to check whether the data arrived unchanged.
- **Stop-and-Wait flow control:** the sender sends one frame and then waits for an acknowledgement before sending the next — simple, but slow.
- **Sliding Window flow control:** the sender is allowed to send several frames before needing an acknowledgement, greatly improving efficiency — used by TCP.

(d) Multiplexing Techniques (Sharing One Channel Among Many Users)

Technique	Full Form	How It Works (Simple Idea)
FDM	Frequency Division Multiplexing	The available frequency range is divided into smaller bands; each conversation/channel gets its own band, all sent at the same time (like many radio stations broadcasting simultaneously on different frequencies).
TDM	Time Division Multiplexing	Each conversation/channel gets the entire frequency, but only for a short time slot, taking turns in rapid rotation (like several people sharing one phone line by each speaking only during their short allotted turn).
WDM	Wavelength Division Multiplexing	Used specifically in fibre optics — different signals are sent as different wavelengths (colours) of light down the same fibre at the same time.

(e) Connection-Oriented vs Connectionless Services

Basis	Connection-Oriented Service	Connectionless Service
Concept	A connection (path) is first established, used, then released — like a telephone call	No prior connection is set up; each packet travels independently — like postal mail
Order of delivery	Data arrives in the same order it was sent	Packets may arrive out of order
Reliability	Generally reliable (acknowledgements used)	May be reliable or unreliable
Example	TCP	UDP, IP

(f) Services vs Protocols

A service is a set of operations that a layer provides to the layer above it (WHAT the layer does); it says nothing about how requests are implemented. A protocol is a set of rules governing the format and meaning of packets/messages exchanged by peer entities within a layer (HOW the service is implemented). This is similar to a restaurant menu versus a chef's recipe: the menu (service) tells the customer what dishes are available; the recipe (protocol) is how the chef actually prepares each dish. A customer can order from the menu without ever knowing the recipe.

★ Quick Recap – 1.4

- Layers simplify design; each layer offers a service to the layer above via an interface, using a protocol to talk to its peer.
- Key design issues: addressing, error control, flow control, multiplexing, and routing.
- FDM splits by frequency, TDM splits by time, WDM splits by wavelength (colour of light in fibre).
- Service = what a layer offers (the menu); Protocol = the rules used to implement it (the recipe).

1.5 Reference Models

A reference model is a conceptual framework that describes the functions a networking system should perform, without specifying exactly how. The two most important reference models are the OSI model and the TCP/IP model.

(a) The OSI Reference Model

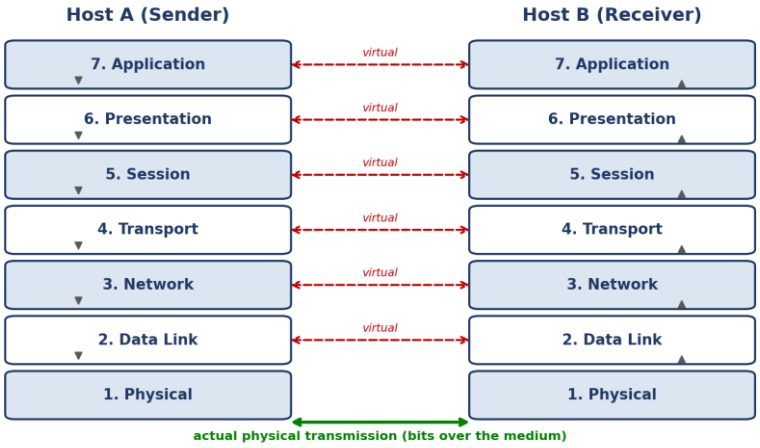
OSI stands for Open Systems Interconnection. It was developed by ISO (International Standards Organization) as a first step towards international standardisation. It has 7 layers, each with a well-defined function.

Layer No.	Layer Name	Main Function
7	Application	Provides network services directly to end-user applications (e-mail, file transfer, browsing)
6	Presentation	Data translation, encryption/decryption, and compression (formatting so both ends understand)
5	Session	Establishes, manages, and terminates sessions (dialogue control) between two communicating hosts
4	Transport	End-to-end delivery of complete messages; error control and flow control between source and destination
3	Network	Routing of packets from source to destination across multiple networks; logical (IP) addressing
2	Data Link	Organises bits into frames; physical/MAC addressing; error detection at the link level
1	Physical	Transmission of raw bits over the physical medium (cables, connectors, voltages, radio signals)

Memory tip (top to bottom): "All People Seem To Need Data Processing" = Application, Presentation, Session, Transport, Network, Data Link, Physical.

How Do the Layers Actually Communicate?

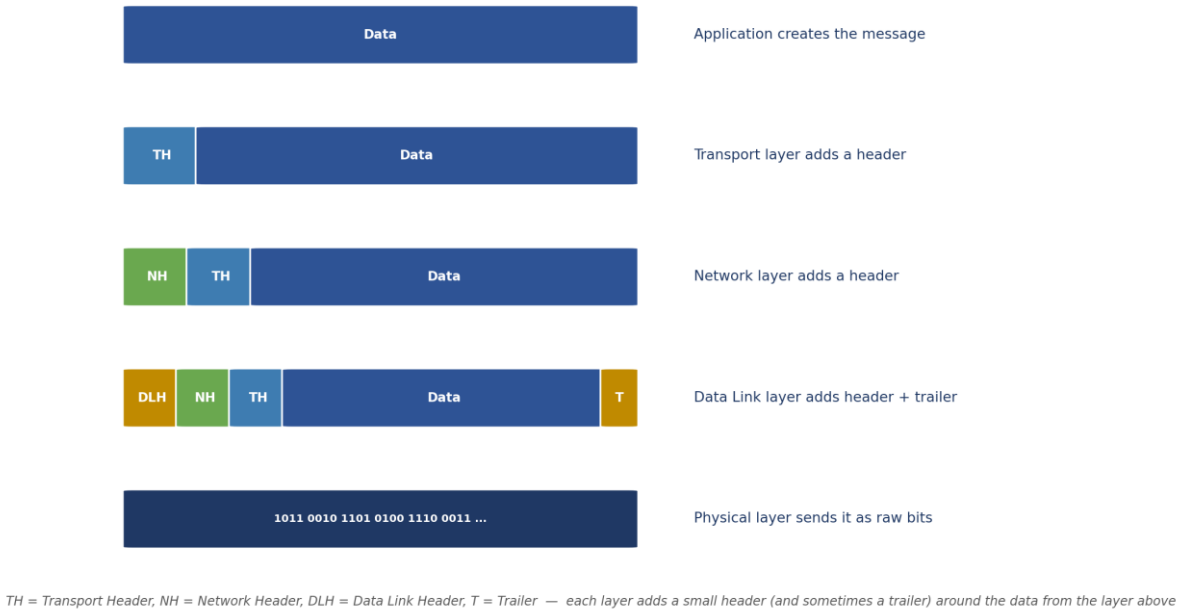
Each layer on the sending machine appears to talk directly to the same layer on the receiving machine (this imaginary, direct conversation is called "virtual communication" or "peer-to-peer communication", shown by the dashed lines below). In reality, data only physically travels down through all the layers on the sender's side, across the wire at the Physical layer, and back up through all the layers on the receiver's side.



Virtual (peer-to-peer) communication vs the actual physical path data takes

Encapsulation — How Data Is Wrapped at Each Layer

As data travels down through the layers on the sending side, each layer adds its own header (a small block of control information) in front of the data — and sometimes a trailer at the end. This is called encapsulation. It works exactly like posting a letter: you write the letter (Data), put it in an inner envelope addressed to the department (Transport Header), that goes inside another envelope addressed to the building (Network Header), and finally inside a courier packet with the delivery address label (Data Link Header). Each "envelope" is opened and removed by the matching layer on the receiving side, in exactly the reverse order.



Encapsulation: each layer wraps the data from the layer above with its own header

(b) The TCP/IP Reference Model

TCP/IP was developed for the ARPANET (the ancestor of the Internet) by the US Department of Defense. It is a simpler, 4-layer model that is the actual foundation of today's Internet.

TCP/IP Layer	Main Function	Common Protocols
Application	Combines OSI's Application, Presentation and Session layers — handles high-level protocols and user services	HTTP, FTP, SMTP, DNS
Transport	End-to-end communication and reliability between source and destination hosts	TCP, UDP
Internet	Routing of packets (datagrams) across networks; corresponds to OSI's Network layer	IP, ICMP, ARP
Host-to-Network (Link)	Combines OSI's Data Link and Physical layers — how bits are sent over the actual medium	Ethernet, Wi-Fi (802.11)

(c) OSI vs TCP/IP — Comparison

Basis	OSI Model	TCP/IP Model
Number of layers	7 layers	4 layers
Approach	Model first, then protocols were designed — more theoretical	Protocols came first (already used on ARPANET); model was described afterward — more practical
Layer separation	Session and Presentation are separate layers	Session and Presentation are merged into the Application layer
Usage today	Used mainly as a teaching/reference tool	Used in actual implementation — it is the basis of the real Internet
Reliability	Guarantees reliable delivery in the Network layer design	Network layer (IP) is connectionless/unreliable; reliability is added at the Transport layer (TCP) only

(d) The Internet

The Internet is a huge "network of networks" that uses the TCP/IP protocol suite to connect millions of smaller academic, business, government, and home networks worldwide.

- It began as ARPANET (1969), an experimental network funded by the US Department of Defense.
- It grew through NSFNET (connecting universities) and later opened up for commercial and public use in the 1990s.
- Today it is composed of end systems (hosts), Internet Service Providers (ISPs — companies that provide your Internet connection), routers, and communication links, all using IP addressing (a unique numeric address for every device, such as 192.168.1.1) and domain names (easy-to-remember names like www.example.com, translated into IP addresses by the DNS — Domain Name System) to communicate.
- Key services running over the Internet include the World Wide Web (WWW), e-mail, file transfer, video streaming, and VoIP.

(e) Frame Relay

Frame Relay is a wide area network (WAN) protocol that operates at the Physical and Data Link layers. It was designed to connect LANs across long distances efficiently, using packet-switching technology over virtual circuits.

- Uses Permanent Virtual Circuits (PVCs) or Switched Virtual Circuits (SVCs) to create logical paths between two endpoints over a shared network.
- It is a "lean" protocol — it does minimal error checking and flow control itself, relying on higher layers (and the good quality of modern digital lines) to handle errors, which makes it fast and efficient.
- Uses statistical multiplexing so that bandwidth is shared dynamically among multiple users/data streams, improving efficiency.
- Historically popular in the 1990s–2000s for connecting corporate branch offices; today it has largely been replaced by MPLS, leased lines, and VPNs over the Internet, but it remains an important example of WAN/packet-switching technology for exams.

(f) Switching Techniques — How Data Actually Moves Through a Network

Type	How It Works	Analogy / Example
Circuit Switching	A dedicated physical path is reserved for the entire duration of the communication before data transfer begins	A telephone call — the line is reserved just for you, even during silence

Packet Switching	Data is broken into small packets, each sent independently and possibly via different paths; reassembled at the destination	Frame Relay, and the Internet (IP) — like sending a book chapter by chapter through different couriers, reassembled on arrival
Message Switching	The entire message is sent as one unit and stored temporarily at each intermediate node before being forwarded ("store-and-forward")	Old telegraph systems — like a relay of runners, each carrying the whole message to the next station

★ Quick Recap – 1.5

- OSI = 7-layer theoretical/reference model; TCP/IP = 4-layer model actually used on the real Internet.
- OSI layers (top-down): Application, Presentation, Session, Transport, Network, Data Link, Physical.
- TCP/IP layers (top-down): Application, Transport, Internet, Host-to-Network.
- Layers communicate virtually (peer-to-peer) but data actually flows down, across the wire, and back up.
- Encapsulation = each layer adds its own header (like nested envelopes) as data goes down the stack.
- The Internet is a global network of networks built on TCP/IP; grew out of ARPANET.
- Frame Relay: WAN packet-switching protocol using virtual circuits; fast because it skips heavy error checking.
- Circuit switching reserves a dedicated path (telephone call); packet switching sends independent packets (the Internet).

UNIT 2: THE PHYSICAL LAYER

The Physical Layer is the lowest layer of the OSI model. It is concerned with transmitting raw bits (0s and 1s) over a communication channel. This unit covers the transmission media (guided and wireless) used to carry these signals, and two important real-world telephone-based systems: the PSTN and the mobile telephone system.

2.1 Guided Transmission Media

Guided media (also called "wired" or "bounded" media) physically guide/confine the signal within a solid medium such as a cable. The three main types are twisted pair, coaxial cable, and fibre optics.

(a) Twisted Pair Cable

- Consists of two insulated copper wires, typically about 1 mm thick, twisted together in a spiral. Twisting the wires together cancels out most electrical interference (crosstalk) from nearby pairs — similar to how braided hair holds together better and resists tangling more than loose strands.
- Types: UTP (Unshielded Twisted Pair) — common and cheap; STP (Shielded Twisted Pair) — has extra shielding, more resistant to noise.
- Used for: telephone lines (local loop) and Ethernet LANs (e.g., Cat 5e cables support up to about 1 Gbps, Cat 6 up to about 10 Gbps over short distances).
- Advantages: inexpensive, easy to install and maintain. Disadvantage: lower bandwidth and more susceptible to noise than other media.

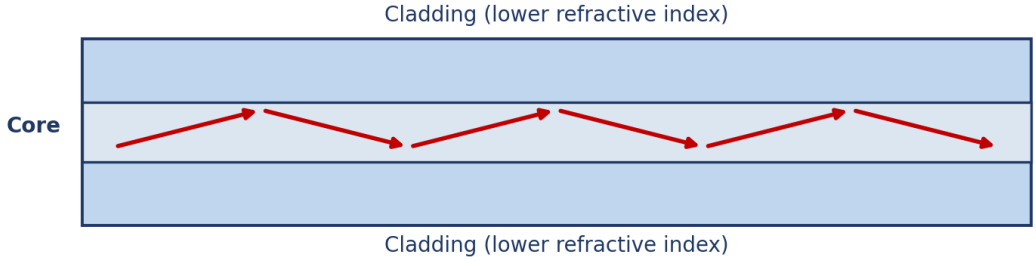
(b) Coaxial Cable

- Has a central copper conductor surrounded by insulation, then a cylindrical metal shield (mesh), and an outer plastic cover.
- The shielding gives coaxial cable better noise immunity and higher bandwidth than twisted pair, allowing it to span longer distances at higher speeds (commonly several hundred Mbps in cable-TV/broadband internet connections).
- Used for: cable television (CATV) networks and, historically, early Ethernet LANs and long-distance telephone lines.

(c) Fibre Optics

A fibre-optic system has three key components: the light source (usually an LED or laser diode), the transmission medium (an ultra-thin glass or plastic fibre), and the detector (a photodiode).

- Works on the principle of total internal reflection — light entering at a suitable angle keeps bouncing inside the glass core and travels along the fibre with very little loss, the way light bounces endlessly down a mirrored tunnel.



Light ray keeps reflecting inside the core (total internal reflection) and travels along the fibre

Total internal reflection keeps the light ray travelling inside the fibre core

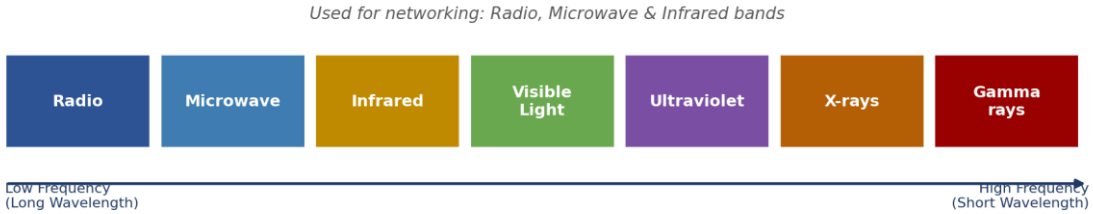
- **Multimode fibre:** has a larger core; multiple light rays ("modes") travel at slightly different angles; cheaper but suitable for shorter distances.
- **Single-mode fibre:** has a very narrow core so light travels almost in a straight line (one mode); used for long-distance, high-speed links (e.g., undersea cables, backbone links) reaching speeds of many Gbps to Tbps.
- Advantages: very high bandwidth, very low signal loss (attenuation), immune to electromagnetic interference, lightweight, and highly secure (very difficult to tap).
- Disadvantages: more expensive, fragile, and requires skilled installation/splicing.

Comparison of Guided Media:

Property	Twisted Pair	Coaxial Cable	Fibre Optic
Bandwidth / Speed	Low–Moderate (up to ~10 Gbps)	Moderate–High (up to ~1 Gbps)	Very High (Gbps to Tbps)
Noise immunity	Low	Moderate	Excellent (immune to EMI)
Cost	Cheapest	Moderate	Most expensive
Typical use	Telephone lines, LAN	Cable TV, older LANs	Backbone, long-distance, high-speed internet

2.2 Wireless Transmission

Wireless (unguided) transmission sends signals through free space (air/vacuum) using electromagnetic waves, without a physical wire. Different parts of the electromagnetic spectrum are used depending on the frequency (radio, microwave, infrared, and light).



The electromagnetic spectrum — networking mainly uses the radio, microwave and infrared bands

(a) Radio Transmission

- Radio waves are easy to generate, can travel long distances, and can penetrate buildings, making them widely used for both indoor and outdoor communication.
- They are omnidirectional — travel in all directions from the source, so the transmitter and receiver do not have to be precisely aligned.
- Used for: FM/AM radio broadcasting, Wi-Fi, Bluetooth, and cordless phones.

(b) Microwave Transmission

- Above 100 MHz, waves travel in nearly straight lines and are highly directional, so transmitter and receiver antennas must be accurately aligned (line-of-sight communication).
- Terrestrial microwave uses tower-mounted parabolic antennas; satellite microwave relays signals via a satellite in orbit, useful for very long-distance communication (e.g., international telephone calls, television broadcast, GPS).
- Disadvantage: does not pass through solid obstacles well and is affected by weather (especially at higher frequencies).

(c) Communication Satellites

Satellites act as mirrors in the sky, receiving a signal from one ground station and relaying it back down to another, far away. They are classified by their orbital height:

Orbit Type	Approx. Altitude	Key Feature	Typical Use
GEO (Geostationary)	~36,000 km	Appears fixed over one point on Earth; only 3 satellites can cover almost the whole globe; but signals take a noticeable ~250 ms delay (high latency)	TV broadcast, weather monitoring
MEO (Medium Earth Orbit)	~5,000–15,000 km	Moderate delay, moderate coverage per satellite	GPS navigation satellites

LEO (Low Earth Orbit)	~500–2,000 km	Very low delay (low latency); needs many satellites working together to cover the Earth	Modern satellite internet (e.g., broadband constellations), some mobile phone systems

(d) Infrared and Millimeter Waves

- Infrared waves are used for short-range communication, e.g., TV remote controls, and older wireless keyboard/mouse/IR ports.
- They do not pass through solid walls, which is actually an advantage for security (a device in one room cannot interfere with a device in the next room).
- Does not require government licensing, unlike radio.

(e) Lightwave Transmission

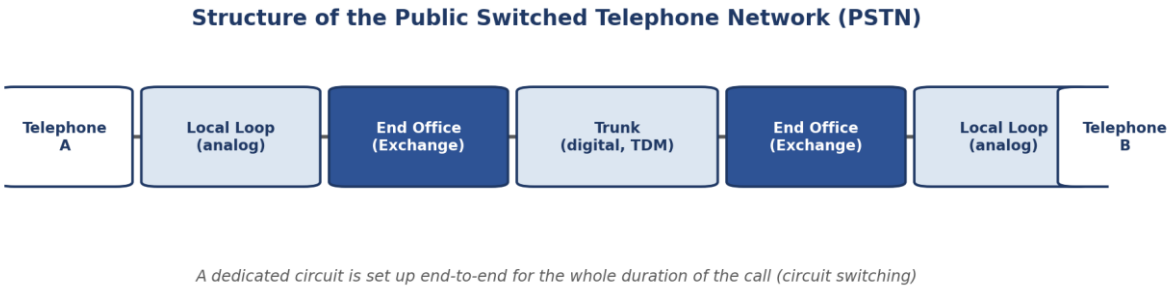
- Unguided laser/light signals can be used to connect networks in nearby buildings by mounting a laser transmitter and receiver on rooftops (line-of-sight optical link).
- Advantages: very high bandwidth and low cost, no licence needed. Disadvantage: narrow beam is greatly affected by weather (fog, heavy rain) and by small vibrations/misalignment of the building.

★ Quick Recap – 2.1 & 2.2

- Guided media (wired): Twisted Pair < Coaxial < Fibre Optic, in order of increasing bandwidth and cost.
- Wireless media: Radio (omnidirectional, penetrates walls) → Microwave (line-of-sight, directional) → Infrared/Light (very short range, blocked by walls).
- Fibre optics uses total internal reflection and offers the highest bandwidth with immunity to electrical noise.
- Satellites: GEO (far, fixed position, high delay) — MEO (GPS) — LEO (close, low delay, needs many satellites).

2.3 The Public Switched Telephone Network (PSTN)

The PSTN is the traditional circuit-switched telephone network that was originally built to carry analog voice calls, and today also carries data. It consists of three main components.



The structure of the PSTN, connecting two telephones end-to-end

(a) Local Loop

- The analog twisted-pair wire connecting a subscriber's telephone (or modem) to the nearest telephone exchange, also called the "last mile".
- Because it was designed for voice, a modem is needed to convert digital computer data into analog signals for transmission over the local loop (and back again at the other end).
- Modern broadband technologies such as DSL (Digital Subscriber Line) use the same existing copper telephone wires to also deliver high-speed internet, by using frequencies above the range needed for voice calls.

(b) Trunks and Digitisation

- High-capacity digital links (fibre optic, microwave, or satellite) that carry many calls simultaneously between telephone exchanges/switching offices.
- Multiplexing (Time Division Multiplexing — TDM) is used to combine thousands of individual calls onto a single high-capacity trunk line (for example, a standard T1 trunk carries 24 voice channels together).
- **PCM (Pulse Code Modulation):** the method used to convert an analog voice signal into digital form for the trunks. In simple terms, the voice wave is measured (sampled) thousands of times per second, each measurement is rounded to the nearest available value (quantised), and then written as a binary number (encoded) — much like taking rapid snapshots of a wave and recording the height of each snapshot as a number.

(c) Switching Offices (Exchanges)

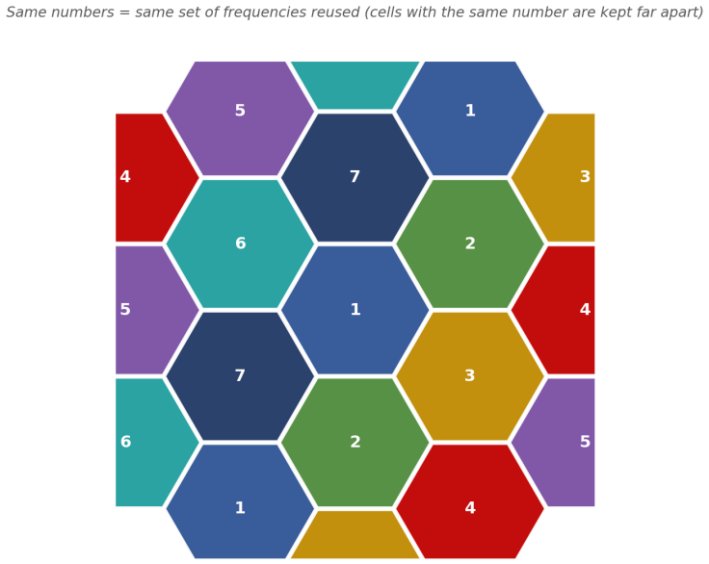
- Telephone calls are connected using circuit switching — a dedicated physical path is set up between caller and receiver for the entire duration of the call, and released when the call ends.
- Local exchanges connect subscribers in a local area; these exchanges are further linked to toll/long-distance exchanges to route calls across cities and countries.

★ Quick Recap – 2.3

- PSTN = Local Loop (analog, to home) + Trunks (digital, high capacity, multiplexed) + Switching Offices (exchanges).
- Uses circuit switching — a dedicated path is reserved for the whole call.
- PCM converts analog voice to digital: sample → quantise → encode.
- DSL reuses the same copper telephone wire to also provide broadband internet.

2.4 The Mobile Telephone System

Mobile telephone systems allow users to communicate while moving, using a technique called cellular communication. The geographic area is divided into small regions called cells, each served by a base station (tower).



Frequency reuse: cells with the same number use the same set of frequencies, kept far apart to avoid interference

- **Frequency reuse:** the same set of frequencies can be reused in cells that are far enough apart, allowing the network to serve many more users than the available frequency bands would otherwise allow. This is exactly why the diagram above shows the same numbers repeating, but never in neighbouring cells.

Handoff (Handover) — Step by Step

As a user moves from one cell to another during a call, the network must transfer the call smoothly:

1. The signal strength from the current base station starts to weaken as the user moves toward the edge of the cell.
2. The mobile phone and the network continuously monitor the signal strength of nearby base stations.
3. The network identifies a neighbouring cell with a stronger signal and reserves a free channel there.
4. The ongoing call is transferred (handed off) to the new base station and channel.
5. The switch happens in a fraction of a second, so the user normally does not notice any interruption.

The mobile system has evolved through several "generations":

Generation	Approx. Era	Key Feature	Technology Example
1G	1980s	Analog voice calls only; poor security, low capacity	AMPS
2G	1990s	Digital voice; better quality, security, and capacity; basic SMS	GSM, CDMA
3G	2000s	Digital data services added — mobile internet, video calling	UMTS, CDMA2000
4G	2010s	All-IP network; very high-speed mobile broadband, HD video streaming	LTE, LTE-Advanced
5G	2020s–present	Ultra-high speed, very low latency, supports IoT and smart devices	5G NR

★ Quick Recap – 2.4

- Mobile systems use cells + base stations; handoff transfers an ongoing call between cells as the user moves, in 5 quick steps.
- Frequency reuse lets the same frequencies serve many users across different (non-adjacent) cells.
- Generations: 1G (analog voice) → 2G (digital voice) → 3G (mobile data) → 4G (high-speed all-IP) → 5G (ultra-fast, low latency).

GLOSSARY OF KEY TERMS

A quick A–Z reference of the important terms used in these notes, explained in plain language.

Term	Meaning
Bandwidth	The maximum amount of data that can be sent through a connection in a given time — like the width of a pipe carrying water.
Bit	The smallest unit of data — either a 0 or a 1.
Circuit Switching	A dedicated communication path is reserved for the whole duration of a call, as in a normal telephone call.
Client–Server Model	An arrangement where powerful server computers provide services/data to many client computers that request them.
Encapsulation	The process of wrapping data with a header (and sometimes a trailer) as it passes down through each network layer.
Frame	A unit of data at the Data Link layer, including addressing and error-checking information.
Frequency Reuse	Using the same radio frequencies in cells that are far apart, to serve more mobile users with limited frequency bands.
Handoff / Handover	The process of transferring an ongoing mobile call from one cell/base station to another as the user moves.
IP Address	A unique numeric address assigned to every device on a network so it can be identified and reached.
ISP	Internet Service Provider — a company that provides an Internet connection to homes/businesses.
LAN	Local Area Network — a network covering a small area such as one building or campus.
Latency	The delay between sending and receiving data.
Modem	A device that converts digital data to analog signals (modulation) and back (demodulation), for transmission over telephone lines.
Multiplexing	Sharing one physical channel among several independent signals/users.
OSI Model	A 7-layer theoretical reference model describing how network communication should be organised.
Packet	A small unit of data, with an address, sent independently through a packet-switched network.
Packet Switching	Data is broken into packets, each routed independently through the network and reassembled at the destination.
Protocol	A set of rules that governs how two devices communicate.
Router	A device that connects different networks and forwards data toward its correct destination network.

Switch	A device that connects computers within a LAN and forwards data only to the intended recipient.
TCP/IP Model	A simpler 4-layer model that is the actual basis of how the real Internet works.
Topology	The physical or logical layout/pattern in which network devices are connected (bus, star, ring, mesh).
Trunk	A high-capacity communication line carrying many multiplexed calls between telephone exchanges.
Twisted Pair	A cable made of two insulated copper wires twisted together to reduce interference; used in telephone lines and LANs.
Virtual Circuit	A logical (not physical) dedicated path set up through a packet-switched network, used in technologies like Frame Relay.
VoIP	Voice over Internet Protocol — making phone calls by sending voice as data packets over the Internet.
WAN	Wide Area Network — a network spanning a large geographical area such as a country or continent.

OVERALL SUMMARY — QUICK REVISION SHEET

Unit 1: Introduction

1. A computer network connects autonomous computers to share resources and information.
2. Goals: resource sharing, high reliability, cost saving, scalability, and communication.
3. Applications: business, home, mobile users, plus related social issues.
4. Hardware classified by transmission technology (broadcast vs point-to-point), scale (PAN, LAN, MAN, WAN, Internetwork), topology (bus, star, ring, mesh), and devices (hub, switch, bridge, router, gateway, modem).
5. Software is organised in layers; each layer offers a service via an interface and talks to its peer using a protocol. Key design issues: addressing, error control, flow control, multiplexing (FDM/TDM/WDM), and routing.
6. OSI model: 7 layers (Application, Presentation, Session, Transport, Network, Data Link, Physical). Data is encapsulated with a header at each layer going down the stack.
7. TCP/IP model: 4 layers (Application, Transport, Internet, Host-to-Network) — the real basis of the Internet.
8. The Internet grew from ARPANET into today's global network of networks using TCP/IP, IP addresses, and domain names.
9. Frame Relay: an efficient WAN packet-switching protocol using virtual circuits; circuit switching reserves a dedicated path while packet switching sends independent packets.

Unit 2: The Physical Layer

1. Guided media: Twisted Pair, Coaxial Cable, Fibre Optic (increasing order of bandwidth and cost); fibre uses total internal reflection.
2. Wireless media: Radio, Microwave, Infrared, and Lightwave transmission, each with different range/directionality trade-offs; satellites are GEO, MEO, or LEO depending on altitude.
3. PSTN = Local Loop + Trunks (using PCM and TDM) + Switching Offices, using circuit switching; DSL reuses telephone wires for broadband.
4. Mobile telephone system uses cells, base stations, handoff, and frequency reuse; has evolved from 1G to 5G.

End of Notes — All the best for your exams